

Event Management System Database Design Document

Document: Event Management System - Database Design
Version: 1.0
Phase: Phase 1 Core Event Operations
Date: June 2026
Status: Draft

Overview

This document defines a complete database design for an Event Management System project intended to manage event discovery, organizer operations, venue scheduling, ticket booking, payments, and notifications. The structure follows a modular approach similar to a production-ready academic or startup project document, with clear separation of master data, operational transactions, and user interactions.

The proposed Phase 1 scope covers five major modules:

- Users and Authentication
- Organizers and Venues
- Event Catalogue
- Ticketing and Bookings
- Payments and Notifications

Phase 2 can later add analytics dashboards, refund workflows, vendor management, QR-based on-site check-in, loyalty systems, and third-party integrations.

Legend for key types

Key	Meaning
PK	Primary Key - unique identifier for every row
FK	Foreign Key - references a row in another table
UK	Unique Key - value must be unique across rows

Module 1 - Users and Authentication

This module manages all users of the platform, including attendees, organizers, and administrators. A single central user table supports role-based access, while supporting tables manage sessions and password recovery.

1.1 USERS

Central user table. The `role` field differentiates attendees, organizers, and admins.

Field name	Data type	Key	Description
id	UUID	PK	Unique identifier for each user
firstname	VARCHAR		User first name
lastname	VARCHAR		User last name
email	VARCHAR	UK	Registered email address
mobile	VARCHAR	UK	Registered mobile number
passwordhash	VARCHAR		Encrypted password hash
role	ENUM		User role: attendee, organizer, admin
city	VARCHAR		User city
state	VARCHAR		User state
status	ENUM		Account status: active, suspended, pendingverification
createdat	TIMESTAMP		Account creation timestamp
updatedat	TIMESTAMP		Last profile update timestamp

1.2 USERSESSIONS

Tracks active login sessions across devices.

Field name	Data type	Key	Description
id	UUID	PK	Unique session identifier
userid	UUID	FK	References <u>USERS.id</u>
token	VARCHAR	UK	Unique session token
ipaddress	VARCHAR		Device IP address
useragent	TEXT		Browser or app information
expiresat	TIMESTAMP		Session expiry time
createdat	TIMESTAMP		Session creation timestamp

1.3 PASSWORDRESETS

Stores one-time password reset requests.

Field name	Data type	Key	Description
id	UUID	PK	Unique reset request identifier
userid	UUID	FK	References <u>USERS.id</u>

Field name	Data type	Key	Description
token	VARCHAR	UK	Password reset token
used	BOOLEAN		TRUE after successful password reset
expiresat	TIMESTAMP		Token expiry time
createdat	TIMESTAMP		Request creation timestamp

Module 2 - Organizers and Venues

This module stores organizer business profiles, verification documents, and venue details. It allows the platform to support multiple event organizers and venue-based scheduling.

2.1 ORGANIZERS

Stores organizer profile and company information.

Field name	Data type	Key	Description
id	UUID	PK	Unique organizer identifier
userid	UUID	FK	References <u>USERS.id</u>
organizationname	VARCHAR		Registered organizer or company name
gstnumber	VARCHAR	UK	GST number for billing and compliance
contactphone	VARCHAR		Organizer contact number
websiteurl	VARCHAR		Official website or portfolio link
address	TEXT		Office address
city	VARCHAR		Organizer city
state	VARCHAR		Organizer state
status	ENUM		pendingverification, active, rejected, suspended
approvedat	TIMESTAMP		Admin approval timestamp
createdat	TIMESTAMP		Organizer registration timestamp

2.2 ORGANIZERDOCUMENTS

Stores uploaded verification documents for organizer approval.

Field name	Data type	Key	Description
id	UUID	PK	Unique document record identifier
organizerid	UUID	FK	References <u>ORGANIZERS.id</u>
doctype	ENUM		gstcertificate, pancard, addressproof, businesslicense
docurl	VARCHAR		Uploaded document URL

Field name	Data type	Key	Description
verificationstatus	ENUM		pending, verified, rejected
uploadedat	TIMESTAMP		Upload timestamp

2.3 VENUES

Stores venues where events are hosted.

Field name	Data type	Key	Description
id	UUID	PK	Unique venue identifier
organizerid	UUID	FK	References <u>ORGANIZERS.id</u>
venueName	VARCHAR		Name of venue or property
venueType	ENUM		auditorium, banquet, openair, conferencehall, stadium, club
address	TEXT		Full venue address
city	VARCHAR		Venue city
state	VARCHAR		Venue state
pincode	CHAR(6)		Venue PIN code
capacity	INT		Maximum attendee capacity
latitude	DECIMAL		Geo latitude
longitude	DECIMAL		Geo longitude
parkingavailable	BOOLEAN		TRUE if parking is available
status	ENUM		active, inactive, maintenance
createdat	TIMESTAMP		Venue creation timestamp

2.4 VENUEIMAGES

Stores venue images for listing and event pages.

Field name	Data type	Key	Description
id	UUID	PK	Unique image record identifier
venueid	UUID	FK	References <u>VENUES.id</u>
imageurl	VARCHAR		Image storage URL
imagetype	ENUM		exterior, interior, stage, seating, parking
sortorder	INT		Display order
createdat	TIMESTAMP		Upload timestamp

Module 3 - Event Catalogue

This module manages categories, event definitions, schedules, media, and pricing structure. It forms the searchable event catalogue used by attendees.

3.1 EVENTCATEGORIES

Stores event category master data.

Field name	Data type	Key	Description
id	UUID	PK	Unique category identifier
categoryname	VARCHAR	UK	Category name such as music, tech, wedding, workshop
description	TEXT		Category description
status	ENUM		active, inactive
createdat	TIMESTAMP		Creation timestamp

3.2 EVENTS

Core event table.

Field name	Data type	Key	Description
id	UUID	PK	Unique event identifier
organizerid	UUID	FK	References <u>ORGANIZERS.id</u>
categoryid	UUID	FK	References <u>EVENTCATEGORIES.id</u>
venueid	UUID	FK	References <u>VENUES.id</u>
eventtitle	VARCHAR		Public event title
eventslug	VARCHAR	UK	SEO-friendly unique slug
shortdescription	TEXT		Brief event summary
fulldescription	LONGTEXT		Full event details
language	VARCHAR		Primary language of event
agerestriction	VARCHAR		Age eligibility details
coverimageurl	VARCHAR		Main event banner image
status	ENUM		draft, pendingapproval, published, cancelled, completed
createdat	TIMESTAMP		Event creation timestamp
updatedat	TIMESTAMP		Last update timestamp

3.3 EVENTSCHEduLES

Stores date and time schedules for events. One event can have multiple sessions or show timings.

Field name	Data type	Key	Description
id	UUID	PK	Unique schedule identifier
eventid	UUID	FK	References <u>EVENTS.id</u>
sessiontitle	VARCHAR		Optional session or show title
startdatetime	DATETIME		Event start timestamp
enddatetime	DATETIME		Event end timestamp
gatesopenat	DATETIME		Entry opening time
ticketbookingdeadline	DATETIME		Last booking cutoff time
status	ENUM		scheduled, rescheduled, cancelled, completed
createdat	TIMESTAMP		Schedule creation timestamp

3.4 EVENTIMAGES

Stores supporting event media gallery.

Field name	Data type	Key	Description
id	UUID	PK	Unique image record
eventid	UUID	FK	References <u>EVENTS.id</u>
imageurl	VARCHAR		Event image URL
imagetype	ENUM		banner, gallery, speaker, stage, poster
sortorder	INT		Display sequence
createdat	TIMESTAMP		Upload timestamp

3.5 TICKETTYPES

Defines pricing slabs for each event schedule.

Field name	Data type	Key	Description
id	UUID	PK	Unique ticket type identifier
scheduleid	UUID	FK	References <u>EVENTSCHEduLES.id</u>
ticketname	VARCHAR		Ticket type such as General, VIP, Premium
price	DECIMAL		Base ticket price
totaltickets	INT		Total inventory allocated

Field name	Data type	Key	Description
ticketsavailable	INT		Remaining available tickets
perks	TEXT		Benefits included with this ticket
sale_start	DATETIME		Ticket sales start time
sale_end	DATETIME		Ticket sales end time
status	ENUM		active, soldout, inactive
createdat	TIMESTAMP		Creation timestamp

Module 4 - Ticketing and Bookings

This module records attendee bookings, issued tickets, seat assignments, and check-in status.

4.1 BOOKINGS

Represents a booking transaction created by an attendee.

Field name	Data type	Key	Description
id	UUID	PK	Unique booking identifier
userid	UUID	FK	References <u>USERS.id</u>
scheduleid	UUID	FK	References <u>EVENTSCHEDULES.id</u>
bookingcode	VARCHAR	UK	Human-readable booking reference
totalamount	DECIMAL		Total payable amount before refund
bookingstatus	ENUM		pending, confirmed, cancelled, refunded
bookedat	TIMESTAMP		Booking creation time
confirmedat	TIMESTAMP		Confirmation time after payment

4.2 BOOKINGITEMS

Stores line items inside a booking for each selected ticket type.

Field name	Data type	Key	Description
id	UUID	PK	Unique booking item identifier
bookingid	UUID	FK	References <u>BOOKINGS.id</u>
tickettypeid	UUID	FK	References <u>TICKETTYPES.id</u>
quantity	INT		Number of tickets booked
unitprice	DECIMAL		Price per ticket at booking time
subtotal	DECIMAL		quantity x unitprice
createdat	TIMESTAMP		Line item creation timestamp

4.3 ISSUEDTICKETS

Stores each issued ticket with unique QR or barcode reference.

Field name	Data type	Key	Description
id	UUID	PK	Unique ticket record identifier
bookingitemid	UUID	FK	References <u>BOOKINGITEMS.id</u>
ticketnumber	VARCHAR	UK	Unique ticket number
attendee_name	VARCHAR		Ticket holder name
attendee_email	VARCHAR		Ticket holder email
seatlabel	VARCHAR		Seat number or zone label, nullable
qrcodevalue	VARCHAR	UK	Unique QR code payload
checkinstatus	ENUM		notcheckedin, checkedin, cancelled
checkedinat	TIMESTAMP		Entry scan timestamp
createdat	TIMESTAMP		Ticket issue timestamp

4.4 EVENTREVIEWS

Stores attendee feedback after event completion.

Field name	Data type	Key	Description
id	UUID	PK	Unique review identifier
userid	UUID	FK	References <u>USERS.id</u>
eventid	UUID	FK	References <u>EVENTS.id</u>
rating	INT		Rating from 1 to 5
reviewtext	TEXT		Optional written feedback
status	ENUM		visible, hidden, reported
createdat	TIMESTAMP		Review submission timestamp

Module 5 - Payments and Notifications

This module stores payment attempts, successful transactions, coupon usage, and notification logs.

5.1 COUPONS

Stores promotional coupon codes.

Field name	Data type	Key	Description
id	UUID	PK	Unique coupon identifier

Field name	Data type	Key	Description
couponcode	VARCHAR	UK	Public coupon code
discounttype	ENUM		percentage, fixedamount
discountvalue	DECIMAL		Coupon discount amount or percentage
minbookingamount	DECIMAL		Minimum eligible amount
maxdiscountcap	DECIMAL		Maximum discount for percentage coupons
validfrom	DATETIME		Coupon start time
validuntil	DATETIME		Coupon end time
usage_limit	INT		Maximum total usages
status	ENUM		active, expired, inactive
createdat	TIMESTAMP		Creation timestamp

5.2 PAYMENTS

Stores payment transaction data for bookings.

Field name	Data type	Key	Description
id	UUID	PK	Unique payment identifier
bookingid	UUID	FK	References <u>BOOKINGS.id</u>
paymentgateway	VARCHAR		Payment provider name
gatewaytransactionid	VARCHAR	UK	Gateway transaction reference
amountpaid	DECIMAL		Paid amount
paymentmethod	ENUM		upi, card, netbanking, wallet
paymentstatus	ENUM		initiated, success, failed, refunded
paidat	TIMESTAMP		Payment completion timestamp
createdat	TIMESTAMP		Payment attempt timestamp

5.3 REFUNDREQUESTS

Stores refund requests against cancelled bookings.

Field name	Data type	Key	Description
id	UUID	PK	Unique refund request identifier
bookingid	UUID	FK	References <u>BOOKINGS.id</u>
paymentid	UUID	FK	References <u>PAYMENTS.id</u>
reason	TEXT		Refund reason

Field name	Data type	Key	Description
refundamount	DECIMAL		Requested or approved amount
status	ENUM		requested, approved, rejected, processed
requestedat	TIMESTAMP		Refund request time
processedat	TIMESTAMP		Refund completion time

5.4 NOTIFICATIONS

Stores in-app, email, and SMS notification logs.

Field name	Data type	Key	Description
id	UUID	PK	Unique notification identifier
userid	UUID	FK	References <u>USERS.id</u>
title	VARCHAR		Notification title
message	TEXT		Notification content
type	ENUM		booking, payment, eventupdate, reminder, account, refund
channel	ENUM		email, sms, inapp
isread	BOOLEAN		TRUE after user reads the notification
sentat	TIMESTAMP		Notification dispatch time

Relationships Summary

The following table summarizes the major foreign key relationships across the system.

From table	Relationship	To table	Meaning
USERSESSIONS	many-to-one	USERS	Each session belongs to one user
PASSWORDRESETS	many-to-one	USERS	Each reset request belongs to one user
ORGANIZERS	one-to-one	USERS	Each organizer maps to one login account
ORGANIZERDOCUMENTS	many-to-one	ORGANIZERS	Each document belongs to one organizer
VENUES	many-to-one	ORGANIZERS	Each venue is managed by one organizer
VENUEIMAGES	many-to-one	VENUES	Each image belongs to one venue
EVENTS	many-to-one	ORGANIZERS	Each event is created by one organizer
EVENTS	many-to-one	EVENTCATEGORIES	Each event belongs to one category
EVENTS	many-to-one	VENUES	Each event is hosted at one venue
EVENTSCHEDULES	many-to-one	EVENTS	Each schedule belongs to one event
EVENTIMAGES	many-to-one	EVENTS	Each image belongs to one event

From table	Relationship	To table	Meaning
TICKETTYPES	many-to-one	EVENTSCHEDULES	Each ticket type belongs to one schedule
BOOKINGS	many-to-one	USERS	Each booking is made by one user
BOOKINGS	many-to-one	EVENTSCHEDULES	Each booking is for one event schedule
BOOKINGITEMS	many-to-one	BOOKINGS	Each line item belongs to one booking
BOOKINGITEMS	many-to-one	TICKETTYPES	Each line item references one ticket type
ISSUEDTICKETS	many-to-one	BOOKINGITEMS	Each issued ticket belongs to one booking item
EVENTREVIEWS	many-to-one	USERS	Each review is submitted by one user
EVENTREVIEWS	many-to-one	EVENTS	Each review is for one event
PAYMENTS	many-to-one	BOOKINGS	Each payment belongs to one booking
REFUNDREQUESTS	many-to-one	BOOKINGS	Each refund request belongs to one booking
REFUNDREQUESTS	many-to-one	PAYMENTS	Each refund request refers to one payment
NOTIFICATIONS	many-to-one	USERS	Each notification is sent to one user

ENUM Values Reference

The following table lists important ENUM fields and their allowed values.

Table	Field	Allowed values
USERS	role	attendee, organizer, admin
USERS	status	active, suspended, pendingverification
ORGANIZERS	status	pendingverification, active, rejected, suspended
ORGANIZERDOCUMENTS	doctype	gstcertificate, pancard, addressproof, businesslicense
ORGANIZERDOCUMENTS	verificationstatus	pending, verified, rejected
VENUES	venuetype	auditorium, banquet, openair, conferencehall, stadium, club
VENUES	status	active, inactive, maintenance
EVENTCATEGORIES	status	active, inactive
EVENTS	status	draft, pendingapproval, published, cancelled, completed
EVENTSCHEDULES	status	scheduled, rescheduled, cancelled, completed
EVENTIMAGES	imagetype	banner, gallery, speaker, stage, poster
TICKETTYPES	status	active, soldout, inactive
BOOKINGS	bookingstatus	pending, confirmed, cancelled, refunded
ISSUEDTICKETS	checkinstatus	notcheckedin, checkedin, cancelled
EVENTREVIEWS	status	visible, hidden, reported
COUPONS	discounttype	percentage, fixedamount

Table	Field	Allowed values
COUPONS	status	active, expired, inactive
PAYMENTS	paymentmethod	upi, card, netbanking, wallet
PAYMENTS	paymentstatus	initiated, success, failed, refunded
REFUNDREQUESTS	status	requested, approved, rejected, processed
NOTIFICATIONS	type	booking, payment, eventupdate, reminder, account, refund
NOTIFICATIONS	channel	email, sms, inapp

Suggested Business Rules

- One organizer can manage multiple venues and multiple events.
- One event can contain multiple schedules, especially for workshops, concerts, or repeated show timings.
- Ticket inventory should decrease only after successful payment confirmation.
- QR code value and ticket number must always remain unique.
- A review should be allowed only for users who have a confirmed booking for the event.
- Refund requests should be allowed only for bookings in eligible cancellation windows.
- Venue capacity and ticket allocations should be validated before publishing event schedules.

Suggested SQL Implementation Notes

- Use UUIDs for distributed-safe primary keys.
- Add indexes on foreign keys, `eventslug`, `bookingcode`, `couponcode`, and `gatewaytransactionid` for faster retrieval.
- Enforce unique constraints on email, mobile, coupon codes, and ticket numbers.
- Use soft-delete or status fields instead of hard deletion for key transactional tables.
- Consider triggers or application-layer logic to reduce `ticketsavailable` after successful payment.
- Store image and document files in cloud object storage and save only URLs in the database.

Conclusion

This Event Management System database design is suitable for an academic major project, internship assignment, or startup prototype. It provides a normalized and modular schema covering user management, organizer workflows, event publishing, ticket sales, payments, and notifications, while keeping enough flexibility for future enhancements such as on-site scanning, advanced analytics, and partner integrations.